Table 2.2 Non-uniform motion

Time (second)	Distance (metre)
0	0
2	1
4	4
6	9
8	16
10	25
12	36

If we plot a graph between the distance travelled and the time taken, it would be as shown in Figure 2.4.

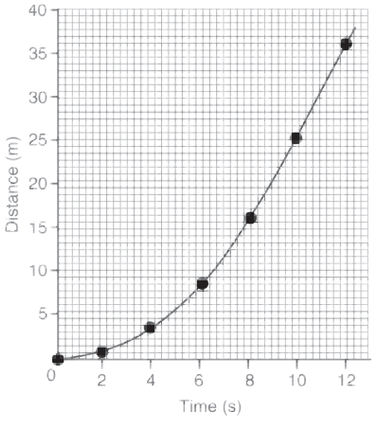


Figure 2.4 The distance time graph for Non-uniform motion

Note that the graph is not a straight line as we got in the case of uniform motion. This nature of the graph shows non–linear variation of the distance travelled by the car with time. Thus, the graph represents motion with non-uniform speed.

2.5.3 Velocity – Time graph

The variation in velocity of an object with time can be represented by velocity – time graph. In the graph, time is represented along the X – axis and the velocity is represented along the Y – axis. If the object moves at uniform velocity, a straight line parallel to X-axis is obtained. This graph shows the velocity – time graph for a car moving with uniform velocity of 40 km/hour.

We know that the product of velocity and time gives displacement of an object moving with uniform velocity. Thus, the area under the velocity – time graph is equal to the magnitude of the displacement. So, the distance (displacement), S covered by the car in a time interval of t can be expressed as,

S = AC × CD

S = Area of the rectangle ABCD (shaded portion in the graph)

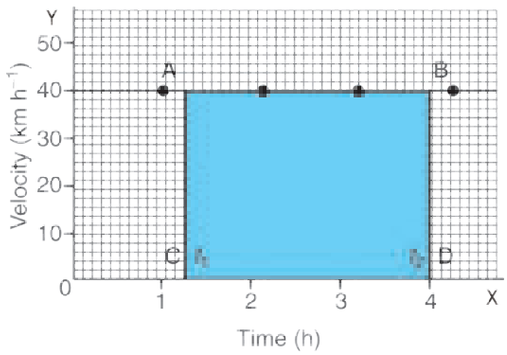


Figure 2.5 Velocity – Time graph

We can also study about uniformly accelerated motion by plotting its velocity – time graph. Consider a car being driven along a straight road. Its velocity for every 5 seconds is noted from the speedometer of the car. The velocity of the car in ms^{-1} at different instants of time is shown in the Table 2.3.

Table 2.3 Uniformly accelarated motion

Time (Second)	Velocity of the Car (ms^{-1})
0	0
5	9
10	18
15	27
20	36
25	45
30	54

In this case, the velocity – time graph for the motion of the car is shown in Figure 2.6 (straight line). The nature of the graph shows that the velocity changes by equal amounts in equal intervals of time. Thus, for all uniformly accelerated motion, the velocity – time graph is a straight line.

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